

OPERATION & MAINTENANCE PUBLICATION

PIG LAUNCHER & RECEIVER

MAINTENANCE MANUAL

MODEL: LSG-PLR

Inspection • Preventive Maintenance • Troubleshooting • Spare Parts



PIPELINE PIGGING EQUIPMENT

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PIG LAUNCHER & RECEIVER

OPERATIONAL AND TECHNICAL MANUAL

PLR-QC SERIES - QUICK OPENING CLOSURE TYPE

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IMPORTANT

Read this manual together with the approved datasheet, GA drawing, P&ID, closure manufacturer instructions, site SOP, permit-to-work and lockout/tagout procedures. Project documents and local regulations take precedence over this generic manual.

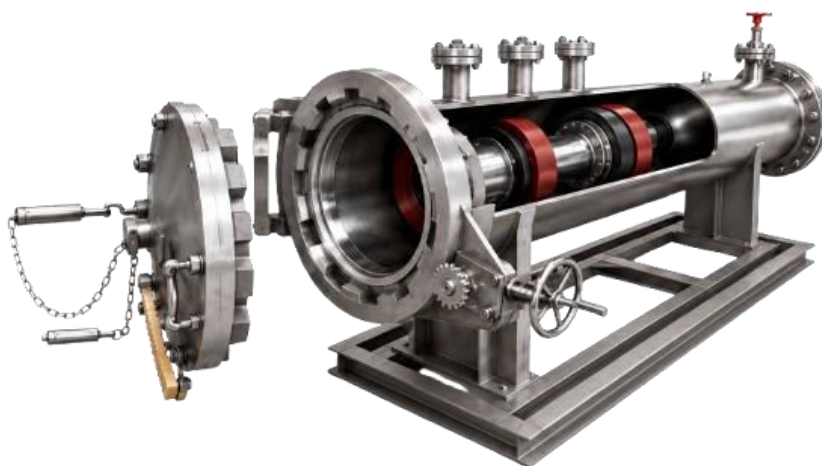
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NOTE: The page numbering and section locations may update when this editable manual is customized for a specific project.



1. Purpose, Scope and Intended Use

1.1 Purpose

This manual provides general technical, installation, commissioning, operating, inspection, maintenance and troubleshooting guidance for horizontal LINE SIG Pig Launchers and Pig Receivers fitted with a quick opening closure. It is intended for trained pipeline operations, mechanical maintenance, inspection and HSE personnel.

1.2 Scope

- Launcher and receiver pressure-containing bodies and reducers.
- Quick opening closure and locking arrangement.
- Vent, drain, kicker/equalizing and isolation connections.
- Pressure indication, pig signaller and related accessories.
- Supports, saddles, lifting lugs and optional skid frame.
- Generic launching and receiving procedures.

1.3 Exclusions

- Pipeline hydraulic design and process simulation.
- Selection or certification of the pig for a specific pipeline.
- Site-specific valve numbering and automated control logic.
- Repair welding or pressure-boundary modification.
- Certified closure design instructions, which remain controlled by the closure manufacturer.

CONTROL

Do not use this manual to override the approved project P&ID, cause-and-effect, operating procedure, closure manual, statutory code or local safety requirements.



2. Safety Requirements and Hazard Controls

2.1 Principal Hazards

Hazard	Potential Consequence	Minimum Control
Stored pressure / trapped gas	Fatal ejection, impact, release of hazardous fluid	Positive isolation, depressurization, venting, draining and independent zero-energy verification
Unexpected valve operation	Repressurization or pig movement	LOTO, valve position control and authorized operator
Toxic / flammable fluid	Exposure, fire or explosion	Gas testing, ventilation, suitable PPE and area controls
Heavy closure / pig	Crush, pinch and manual-handling injury	Mechanical handling aids and controlled swing area
Static electricity	Ignition risk	Bonding/earthing where required by site procedure
Hot or cold service	Burns or cold injury	Temperature verification and service-specific PPE
Pyrophoric deposits / scale	Fire, toxic exposure	Service-specific cleaning and waste handling plan

2.2 Mandatory Non-Bypassable Rules

1. Obtain an approved permit-to-work and follow the site-specific pigging procedure.
2. Identify, isolate, lock and tag every pressure and energy source connected to the trap.
3. Open the vent and drain to a verified safe system and confirm the barrel is fully depressurized.
4. Verify zero pressure using the gauge and an independent means before operating the closure.
5. Stand clear of the closure opening path and never position personnel in front of the closure.
6. Do not force, hammer or heat the closure locking mechanism.
7. Do not bypass a closure pressure-warning device, interlock or safety lock.
8. Stop work if valve positions, pressure condition, pig location or line-up are uncertain.



DANGER

A pressure gauge reading of zero alone is not sufficient. A blocked or isolated gauge can show zero while pressure remains trapped. Verify vent flow has ceased and use the approved independent zero-pressure verification method.

3. Product Description and Operating Principle

A pig launcher is installed at the upstream end of a pipeline section to insert and propel a pig into the line. A pig receiver is installed downstream to decelerate, contain and permit safe removal of the pig after passage. The trap barrel is larger than the pipeline bore to accommodate the pig and is connected to the line through a reducer or neck. Typical LINE SiG pig launcher/receiver arrangement. Actual project configuration may differ.



3.1 Typical Functional Connections

Connection	Function
Main line connection	Connects trap to the pipeline.
Kicker / equalizing line	Introduces or equalizes pressure and provides flow to launch or receive the pig.
Vent	Removes gas and relieves trapped pressure to an approved system.
Drain	Remove liquid, condensate and clean fluid to an approved closed-drain system.
Pressure gauge connection	Local pressure indication.
Pig signaller nozzle	Detects pig passage at the selected location.
Relief / thermal relief connection	Protects blocked-in liquid sections where specified.

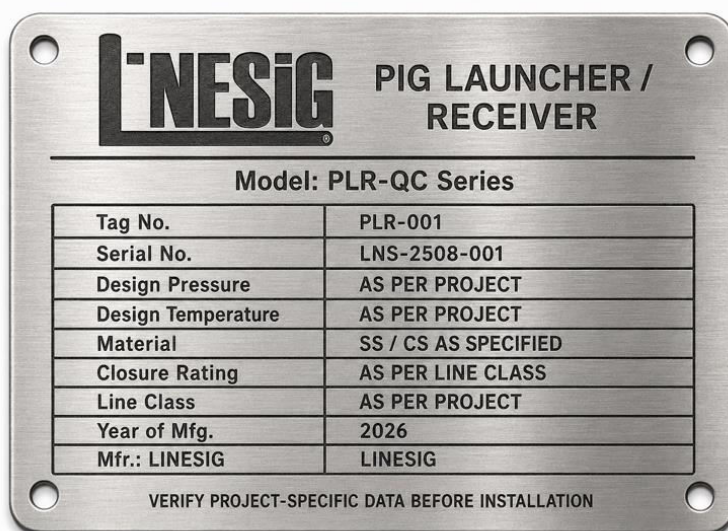
4. Technical Data and Project-Specific Information

PROJECT DATA	Complete this section from the approved equipment datasheet and GA drawing before issuing the manual for operation. Do not assume ratings from a generic product range.
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Parameter	Launcher	Receiver
Tag No.		
Model / Serial No.		
Pipeline Size / Schedule		
Barrel Size / Schedule		
Design Pressure		
Design Temperature		
MAWP / Test Pressure		
Material of Construction		
Corrosion Allowance		
Closure Make / Model		
Design Code		
Service Fluid		
Dry / Operating Weight		
Closure Swing Direction		
Required Pig Length / Diameter		

4.1 Nameplate Verification

- Verify tag number, serial number, design pressure, design temperature and material before installation.
- Confirm closure rating is equal to or greater than the pressure boundary rating.
- Confirm all nozzles, flanges, valves and fittings match the approved line class.



4.2. Identification of Components and Interfaces

Item	Component	Inspection Focus
1	Quick opening closure	Locking segments, hinge, seal, warning device and pressure boundary condition
2	Barrel / trap body	Corrosion, dents, weld condition and cleanliness
3	Reducer / neck	Alignment, wall condition and welds
4	Main line flange / weld end	Rating, gasket surface and alignment
5	Kicker / equalizing connection	Valve condition and flow path
6	Vent connection	Unrestricted safe discharge path
7	Drain connection	Drainability and blockage
8	Pressure gauge	Range, calibration and isolation valve
9	Pig signaller	Trigger movement, reset and leak tightness
10	Supports / saddles / skid	Level, corrosion and anchorage
11	Lifting lugs	Condition and load markings
12	Relief device, if fitted	Set pressure, certification and discharge route

Typical intrusive mechanical pig signaller. Use the project-specific signaller manual for setting and overhaul.

SECTION 5 – MAINTENANCE, TROUBLESHOOTING & SPARES

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Mandatory Safety Rule: Always isolate, depressurize, drain, vent and verify zero pressure before performing inspection, maintenance or component replacement.

5.1 Preventive Maintenance Schedule

Frequency	Maintenance Activities
Before each use	Inspect closure seal; verify pressure gauge condition; inspect pig signaller operation; confirm vent and drain paths are clear; check closure locking mechanism for free movement.
Monthly	Check vent and drain valves; inspect supports and protective coating; inspect accessible fasteners; check for leakage, corrosion and mechanical damage.
Quarterly	Clean accessible internal surfaces; re-lubricate approved moving parts; inspect pig signaller; operate closure mechanism through a full cycle under safe, depressurized conditions.
Annually	Inspect closure locking mechanism; verify pressure gauge calibration; inspect coating and supports; check all accessible fasteners; review maintenance and incident history.
Planned shutdown	Clean internals; inspect seals and gaskets; lubricate moving parts; inspect closure locking components; replace worn or damaged parts as required; perform functional and leak checks before return to service.

Maintenance Inspection Record

Date	Equipment Tag	Activity Performed	Condition / Findings	Action Taken	Inspector

5.2 Troubleshooting

Symptom	Possible Cause	Corrective Action
Leakage at closure	Worn, damaged, displaced or contaminated closure seal; sealing surfaces damaged or dirty.	Isolate and depressurize. Inspect and clean sealing surfaces. Replace seal using the approved seal kit. Verify correct installation and perform leak test.
Closure difficult to open or close	Locking mechanism contaminated, dry, misaligned, corroded or mechanically damaged.	Confirm zero pressure. Clean and inspect mechanism; lubricate only with approved lubricant; correct alignment; replace damaged parts.
Pig signaller not actuating	Trigger mechanism fouled, stuck, damaged, incorrectly set or not adequately inserted.	Isolate and depressurize. Remove and inspect signaller; clean internals; verify free movement, insertion depth and orientation; replace worn parts.
Pig signaller gives false indication	Loose linkage, excessive vibration, incorrect setting or damaged reset mechanism.	Inspect mounting and linkage; tighten fasteners; verify reset and trigger position; replace defective components.
High pressure loss or poor equalization	Blocked equalizing line, partially closed valve, undersized connection, leakage or debris accumulation.	Inspect equalizing path, valves and fittings; clean or flush obstruction; repair leakage; verify valve sequence.
Drain or vent blocked	Debris, scale, corrosion products or frozen/solidified process residue.	Keep unit isolated and depressurized. Open safely, clean or flush the line; inspect and replace blocked valve or strainer if fitted.
External leakage at nozzle or connection	Loose fastener, damaged gasket, cracked fitting or pipe strain.	Depressurize; inspect joint; renew gasket or fitting; tighten to approved torque; remove external piping loads.
Corrosion or coating damage	Environmental exposure, chemical attack, impact damage or coating deterioration.	Clean and assess affected area; repair coating using approved system; evaluate wall condition where corrosion is significant; increase inspection frequency.
Abnormal movement or vibration	Loose support bolts, inadequate foundation, pipe strain or pulsation.	Stop operation if unsafe; inspect supports and anchors; correct foundation or piping alignment; tighten fasteners and investigate pulsation source.

Troubleshooting Record

Date	Symptom	Root Cause	Corrective Action	Verified By	Status

5.3 Recommended Spares

Item	Spare Part	Description / Scope	Recommended Holding
1	Closure Seal Kit	Primary and secondary closure seals, O-rings and approved soft parts as applicable.	Minimum one complete kit per installed unit or project requirement.
2	Pressure Gauge	Gauge of approved range, connection and material according to the equipment datasheet.	One spare per site or critical service requirement.
3	Pig Signaller Internal Kit	Trigger, pivot, spring, seals, reset parts and other serviceable internals as applicable.	One kit for each signaller type installed.
4	Vent / Drain Valves	Ball, needle or other valve type matching the approved specification.	At least one of each critical size/type.
5	Gasket Set	Gaskets for closure interfaces, nozzles and auxiliary connections where applicable.	One complete equipment set.
6	Fastener Set	Approved high-strength bolts, nuts and washers for critical closure or connection locations.	One selected emergency set.
7	Closure Locking Components	Pins, rollers, segments, retaining items or mechanism parts according to closure design.	As recommended by closure manufacturer.
8	Lubricant / Corrosion Preventive	Approved lubricant and preservation compound compatible with seals and process service.	Site maintenance stock.

Spare Parts Identification and Reorder Record

Part No.	Description	Qty. Installed	Qty. in Stock	Minimum Stock	Reorder Required	Remarks

Storage Requirements

Store seals, gaskets and soft parts in sealed packaging away from sunlight, heat, ozone and chemicals. Protect metallic parts from moisture, dust and corrosion; apply approved preservation where necessary. Maintain identification and traceability to equipment tag, model, size, material and revision. Observe shelf-life limits and use first-expiring stock first. Do not substitute materials, dimensions or ratings without engineering approval.



NOTE:

ALWAYS DEPRESSURIZE AND ISOLATE BEFORE MAINTENANCE

6. Receiving, Storage and Preservation

6.1 Receiving Inspection

- Check packing list, tag number, dimensions and accessories against the purchase order.
- Inspect the body, closure, nozzles, supports, coating and instruments for shipping damage.
- Verify flange protectors, desiccants and temporary closures are intact.
- Record damage with photographs before unpacking or repair.

6.2 Storage

- Store on level, load-bearing supports; do not place the pressure body directly on the ground.
- Protect all openings against moisture, dust, animals and foreign objects.
- Maintain internal preservation and renew it at the specified interval.
- Rotate or exercise moving closure parts periodically only when depressurized and under controlled conditions.
- Protect elastomer seals from sunlight, ozone, heat and incompatible chemicals.

Storage Check	Weekly	Monthly	Quarterly
External condition	X		
Weather protection	X		
Flange covers / openings	X		
Internal preservation		X	
Coating / corrosion		X	
Closure movement (if permitted)			X
Desiccant condition		X	

7. Lifting, Handling and Transportation.

LIFTING

Use only engineered lifting points identified on the approved GA drawing. Never lift from the closure, nozzles, valves, instruments, pig signaller or piping.

- Prepare an approved lifting plan based on verified dry or operating weight and center of gravity.
- Use certified slings, shackles and spreader beams of adequate capacity.
- Protect coating and machined surfaces from sling damage.
- Control the closure door and loose items during transport.
- Use tag lines and establish an exclusion zone.
- Set the equipment down on prepared supports without impact.



7.1 Pre-Lift Checklist

Check Point	Verified
Weight and center of gravity confirmed	☐
Lifting lugs inspected	☐
Rigging certificates valid	☐
Closure secured	☐
Loose items removed or restrained	☐
Exclusion zone established	☐
Weather and ground conditions acceptable	☐

8. Installation and Mechanical Completion

8.1 Foundation and Supports

- Install on a level, rigid foundation designed for dead, operating, hydrotest and piping loads.
- Locate supports beneath the designed saddles or skid support points.
- Shim or grout as specified; do not distort the barrel.
- Provide adequate closure swing clearance and safe operator access.

8.2 Piping Connection

- Align piping without forcing the launcher or receiver nozzles.
- Support connected piping independently.
- Verify flange faces are parallel and gasket/bolting match the line class.
- Confirm vent and drain lines slope continuously to the designated safe system.
- Install valves in the direction and orientation shown on the approved P&ID.

8.3 Instrument and Accessory Installation

- Fit pressure gauges with appropriate range and valid calibration.
- Install the pig signaller at the designated nozzle and verify insertion depth.
- Confirm electrical bonding, earthing and hazardous-area requirements where applicable.
- Install relief devices and closure safety interlocks exactly as approved.

PIPE STRAIN

External piping loads can damage the pressure boundary or prevent correct closure sealing. Do not use the trap to pull misaligned piping into position.

9. Pre-Commissioning and Commissioning

9.1 Mechanical Completion Checks

Check Point	Record / Acceptance
Equipment tag and nameplate	Matches approved documents
Internal cleanliness	Free of debris, tools and preservation residue
Closure operation	Smooth, complete lock engagement
Closure seal	Correct type, undamaged and lubricated as approved
Bolting	Correct grade and torque/sequence
Vents and drains	Connected, accessible and clear
Pig signaller	Correct insertion, reset and leak tightness
Pressure gauge	Correct range and valid calibration
Supports and anchors	Complete and level
Coating repairs	Completed and accepted
P&ID line-up	Verified and marked up
Hydrotest / leak test	Approved record available

9.2 Commissioning

9. Confirm the approved commissioning procedure, communication plan and emergency response arrangements.
10. Verify the trap is empty, clean, closed and correctly locked.
11. Pressurize slowly using the designated equalizing or kicker path.
12. Monitor pressure, leakage, abnormal movement and support condition.
13. Hold at the specified operating or leak-test pressure.
14. Depressurize and correct any defect under an approved work permit.
15. Record final valve positions and release the equipment to operations.

10. Pig Selection and Compatibility

SELECTION

The pipeline operator is responsible for selecting a pig compatible with pipeline bore, bends, valves, tees, wall thickness, internal restrictions, service fluid, differential pressure and receiving facilities.

Selection Factor	Required Verification
Pig outside diameter	Appropriate interference with minimum and maximum pipeline ID
Pig length	Fits trap barrel, closure and handling space
Bend capability	Compatible with minimum bend radius and geometry
Valve passage	All valves are full-bore or confirmed passable
Branch connections	Barred or protected where required
Materials	Compatible with service fluid and temperature
Differential pressure	Within pig and pipeline operating limits
Instrumentation / ILI tool	Battery, data, orientation and handling requirements
Receiving energy	Receiver design and operating sequence provide controlled arrival

10.1 Pre-Run Pig Inspection

- Inspect cups/discs, body, magnets, brushes, transmitters and seals.
- Measure and record critical dimensions and weight.
- Confirm tracking frequency and battery status where applicable.
- Verify the pig is clean, correctly assembled and oriented.

11. Operating Philosophy and Valve Sequencing

Valve numbering and sequence shall be taken from the approved P&ID and site procedure. The generic philosophy below is not a substitute for the actual operating procedure.

Functional Valve	Typical Purpose
Main line isolation valves	Isolate the trap from the operating pipeline.
Kicker / equalizing valve	Pressurizes the trap and establishes launching/receiving flow.
Vent valve	Relieves gas or vapor to an approved system.
Drain valve	Removes liquid and confirms depressurization where suitable.
Bypass valve	Maintains line flow or controls differential pressure where designed.

11.1 Universal Safe Sequence

ISOLATE → LOCK/TAG → VENT/DRAIN → VERIFY ZERO ENERGY → OPERATE CLOSURE → LOAD/REMOVE PIG → CLOSE/LOCK → LEAK CHECK → PRESSURIZE SLOWLY → OPERATE VALVES IN APPROVED SEQUENCE



SEQUENCE

Never rely on memory. Use a valve line-up sheet, independent verification and clear communication between control-room and field operators.

12. Pig Launching Procedure

PREREQUISITE

Use only after the procedure has been adapted to the project P&ID, valve numbers, fluid service and emergency arrangements.

16. Obtain authorization, permit-to-work and communication with the control room.
17. Confirm the downstream receiver is prepared and the pipeline route is ready.
18. Identify and close the launcher isolation valves in the approved order.
19. Apply LOTO and verify isolation from every pressure source.
20. Open vent and drain valves to the approved disposal system.
21. Confirm zero pressure by gauge and independent verification.
22. Unlock and open the closure while standing clear of the opening path.
23. Inspect and clean the barrel, closure seal groove and locking components.
24. Inspect, lubricate where applicable, orient and load the pig using suitable handling equipment.
25. Close the closure, engage the locking mechanism fully and verify the safety lock/interlock.
26. Close drain and vent as specified, then equalize and pressurize the launcher slowly.
27. Perform a leak check and stop if leakage or abnormal movement occurs.
28. Open the downstream/mainline path and kicker valve in the approved sequence to launch the pig.
29. Confirm pig departure using the pig signaller and process indicators.
30. Restore the launcher to the specified standby condition and complete the operation record.

12.1 Launch Record

Date/Time	Pig ID	Line Pressure	Signaller Indication	Operator	Remarks

13. Pig Receiving Procedure

31. Confirm the pig has been launched and estimate arrival using approved tracking information.
32. Verify receiver line-up, isolation readiness, drain capacity and safe work zone.
33. Place the receiver in service using the approved valve sequence.
34. Monitor pressure, vibration, sound, pig signaller and tracking system.
35. Confirm pig arrival and allow conditions to stabilize.
36. Isolate the receiver from every live pressure source.
37. Apply LOTO and open vent/drain valves to the approved systems.
38. Wait for drainage and verify zero pressure independently.
39. Stand clear and open the quick opening closure safely.
40. Remove the pig with appropriate handling equipment; treat deposits and fluids as hazardous until classified.
41. Inspect the pig, barrel, seal, closure and signaller trigger.
42. Clean the receiver, replace or lubricate seals as approved, close and lock the closure.
43. Restore the receiver to the specified standby or operating condition.
44. Complete the receiving, inspection and waste records.



ARRIVAL ENERGY

A pig can arrive with significant kinetic energy. Personnel shall not stand in line with the receiver or closure and shall maintain the site-defined exclusion zone during expected arrival.

14. Shutdown, Isolation and Return to Service

14.1 Planned Shutdown

- Notify affected operations and obtain approved isolation plan.
- Stop or divert process flow as required.
- Close isolation valves, apply LOTO and dissipate all stored pressure.
- Drain, purge, wash or inert the trap as required by the service.
- Gas test and maintain ventilation before internal inspection.
-

14.2 Return to Service

45. Confirm work is complete and all personnel, tools and temporary materials are removed.
46. Verify the closure seal and locking system are assembled correctly.
47. Confirm all vents, drains, gauges and signallers are restored.
48. Remove LOTO only under the authorized procedure.
49. Pressurize slowly and conduct a leak check.
50. Restore valve line-up and record return-to-service authorization.

15. Inspection and Preventive Maintenance

Frequency	Minimum Activities
Before each operation	Inspect closure, seal, pressure gauge, vent/drain path, pig signaller, valve position indicators and work area.
Monthly	Inspect coating, supports, anchorage, leakage, corrosion and valve operability.
Quarterly	Clean and lubricate approved moving parts; function-test signaller and closure warning device.
Annually	Detailed closure inspection, gauge calibration, wall-condition review, fastener inspection and relief-device verification.
Major shutdown	Internal examination, NDT/thickness survey as required, seal replacement, valve service and pressure/leak testing after repair.

15.1 Inspection Acceptance

- No active leakage, unacceptable corrosion, cracks, distortion or loose supports.
- Closure locks fully and uniformly without force.
- Seal is free from cuts, swelling, hardening, extrusion or chemical attack.
- Vent and drain paths are unobstructed.
- Pig signaller trigger moves freely and resets positively.
- Pressure gauge and safety devices are within calibration/certification.

16. Quick Opening Closure Maintenance

CLOSURE	The closure manufacturer manual governs disassembly, adjustment, seal selection, torque, allowable wear and replacement parts. Do not modify the closure or locking mechanism.
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16.1 Routine Service

51. Verify the trap is isolated and at zero pressure.
52. Clean the door, hub, locking segments, seal groove and warning device.
53. Inspect for corrosion, galling, mechanical damage and abnormal wear.
54. Inspect the seal and replace with the approved material and size if defective.
55. Apply only the manufacturer-approved lubricant in the specified quantity.
56. Operate the closure through a full cycle and confirm uniform engagement.
57. Inspect hinge pins, supports, handles and safety lock.
58. Record the condition and parts replaced.

16.2 Prohibited Actions

- Grinding, filing or welding locking segments without approved engineering repair.
- Substituting an unapproved seal cross-section or material.
- Using excessive force, hammering or heating.
- Bypassing the pressure warning device or interlock.
- Opening when pressure status is uncertain.

17. Pig Signaller Operation and Maintenance

An intrusive pig signaller uses a trigger extending into the pipe bore. Pig contact rotates or displaces the mechanism and changes the external indicator state. Some versions also provide electrical or wireless output.

17.1 Functional Check

59. Isolate and depressurize the associated trap/nozzle.
60. Confirm the signaller model, insertion length and orientation.
61. Operate the trigger manually using the approved test method.
62. Verify the flag changes state and resets positively.
63. Verify electrical contacts or wireless indication where fitted.
64. Inspect seals, fasteners and leakage path.
65. Return the indicator to the specified armed position.

17.2 Typical Faults

Fault	Likely Cause	Action
No indication	Stuck trigger, incorrect insertion, damaged pig contact	Isolate, remove, clean, inspect and reset; verify dimensions
False indication	Vibration, loose mechanism, incorrect setting	Inspect mounting and internal mechanism
External leakage	Seal or pressure fitting damage	Depressurize and replace approved seal/assembly
Flag does not reset	Contamination or worn spring/linkage	Clean or replace approved internal kit

18. Troubleshooting

Symptom	Possible Cause	Corrective Action
Closure leaks during pressurization	Damaged/dirty seal, incomplete lock engagement, damaged sealing surface	Depressurize, isolate, clean and inspect; replace approved seal; repair only under approved procedure
Closure difficult to open	Residual pressure, contaminated mechanism, misalignment	Stop; verify zero pressure; clean/lubricate; inspect hinge and locking parts
Trap will not pressurize	Vent/drain open, wrong valve line-up, leakage	Check approved valve line-up and leak points
Pig does not launch	Insufficient differential pressure, wrong pig size, bypass leakage, obstruction	Stop and follow stuck-pig contingency plan; do not exceed approved pressure
Pig not received	Tracking uncertainty, bypassed receiver, pig stopped upstream	Verify tracking and line-up; activate approved contingency plan
Excessive vibration/noise	Rapid valve operation, slugging, pig impact, unsupported piping	Stabilize operation, close/open valves gradually, inspect supports
Drain blocked	Debris, wax or scale	Isolate, depressurize and clean using approved method
Corrosion/coating damage	Service or environmental exposure	Assess wall thickness; repair coating; engineering evaluation if metal loss exists



STUCK PIG

Do not improvise by increasing pressure beyond approved operating limits. Use the operator's engineered stuck-pig response plan, tracking data and risk assessment.

19. Recommended Spare Parts and Consumables

Item	Recommended Holding	Notes
Closure primary seal kit	2 sets per closure	Correct material, size and manufacturer part number
Closure secondary seals / O-rings	2 sets	Where applicable
Pig signaller internal kit	1 complete kit per model	Trigger, seals, springs and linkage as applicable
Pressure gauge	1 spare per range/type	Calibrated before installation
Vent and drain valve soft parts	1 service kit per valve type	Service-compatible materials
Flange gaskets	2 sets per size/class	Approved line-class material
Fastener set	As project requirement	Correct grade and coating
Approved lubricant / anti-seize	Minimum site stock	Use only approved products
Coating repair kit	Minimum site stock	Compatible with original coating system

19.1 Spare Parts Identification

Order spares using equipment tag, serial number, closure make/model, signaller model, drawing number and part number. Do not select pressure-boundary or closure parts by visual similarity alone.

20. Preservation, Storage and Decommissioning

20.1 Long-Term Shutdown

- Drain and clean the equipment using a service-compatible method.
- Dry or inert the interior where required.
- Apply approved internal corrosion inhibitor or vapor-phase inhibitor.
- Protect machined surfaces, nozzles and flange faces.
- Store closure seals as recommended or remove them for controlled storage.
- Inspect preservation at documented intervals.

20.2 Decommissioning

66. Confirm permanent isolation and disconnection from all pressure sources.
67. Drain, purge, decontaminate and gas test.
68. Identify hazardous residues, coatings and materials.
69. Prepare a lifting and dismantling plan.
70. Remove instruments, valves and reusable components under controlled conditions.
71. Dispose or recycle materials in accordance with local environmental requirements.
72. Retain the final decommissioning record.

21. Forms, Checklists and Records

21.1 Pre-Operation Safety Checklist

No.	Check Point	Yes	No	N/A	Initial
1	Approved SOP and P&ID available	☐	☐	☐	
2	Permit-to-work approved	☐	☐	☐	
3	Qualified operators assigned	☐	☐	☐	
4	Communication established	☐	☐	☐	
5	Correct pig identified and inspected	☐	☐	☐	
6	Launcher/receiver line-up independently verified	☐	☐	☐	
7	Isolation and LOTO points identified	☐	☐	☐	
8	Vent and drain routes confirmed safe	☐	☐	☐	
9	Pressure gauge functional	☐	☐	☐	
10	Closure and seal inspected	☐	☐	☐	
11	Pig signaller armed/reset	☐	☐	☐	
12	Exclusion zone established	☐	☐	☐	
13	PPE and gas testing requirements met	☐	☐	☐	
14	Emergency response arrangements confirmed	☐	☐	☐	

21.2 Launching / Receiving Operation Record

Record Field	Entry
Date / Shift	
Pipeline / Segment	
Launcher Tag	
Receiver Tag	
Pig ID / Type	
Pig Diameter / Length	
Service Fluid	
Operating Pressure / Temperature	
Launch Time	
Expected Arrival	
Actual Arrival	
Launcher Signaller Indication	
Receiver Signaller Indication	
Control Room Operator	
Field Operator	
Permit No.	
Abnormal Events / Remarks	

21.3 Maintenance and Inspection Record

Field	Entry
Equipment Tag	
Serial No.	
Inspection Date	
Operating Hours / Cycles	
Work Order / Permit	
Inspector / Technician	

Component	Condition / Measurement	Action Required	Completed By
Pressure body			
Closure locking system			
Closure seal and groove			
Hinge and handles			
Vent and drain valves			
Pressure gauge			
Pig signaller			
Supports / anchors			
Coating / corrosion			
Relief device / interlock			

Next inspection due: _____ **Approved by:** _____

21.4 Commissioning / Handover Certificate

Requirement	Status / Reference
Approved datasheet and GA drawing available	
Approved P&ID and valve list available	
Material and fabrication dossier accepted	
Hydrotest / pressure-test records accepted	
NDT reports accepted	
Coating inspection accepted	
Closure certificate and manual available	
Pig signaller data and test records available	
Mechanical completion punch points closed	
Operating and maintenance training completed	
Recommended spares delivered	
Equipment released for operation	

Role	Name	Signature	Date
Construction			
QA/QC			
Operations			
Maintenance			
HSE			
Client / Owner			

22. References and Safety Basis

This generic manual was prepared using the user-approved LINE SIG product concept, established pipeline operating practice and the following public safety sources. Project-specific codes, specifications and approved operating procedures remain controlling.

Reference	Use in This Manual
PHMSA Interpretation PI-83-0101	Emphasizes that injuries and fatalities can result from unsafe launcher/receiver procedures and that closure opening under pressure must be prevented.
PHMSA O&M Enforcement Guidance, Part 195	Supports written procedures for safe pressure relief before pig insertion or removal.
OSHA 29 CFR 1910.147 - Control of Hazardous Energy	Basis for lockout/tagout, isolation, control of stored energy and worker training.
OSHA guidance on multiple energy sources	Supports dissipation or restraint of pneumatic, hydraulic, chemical, thermal and other stored energy.
Approved project datasheet, GA drawing and P&ID	Controls actual equipment ratings, interfaces and valve sequencing.
Closure manufacturer manual	Controls closure-specific operation, inspection, adjustment and repair.

22.1 Final Responsibility Statement

The owner/operator is responsible for converting this manual into a site-specific controlled procedure, completing all project data, validating valve sequences, performing a task risk assessment, training personnel and ensuring compliance with applicable law and company requirements.

FINAL	No launcher or receiver closure shall be opened until the trap is positively isolated, locked/tagged, vented, drained and independently verified to contain no hazardous pressure or stored energy.
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Engineered Flow. Protected Systems. Assured Performance.

Document No.	Revision	Contact
LS-PLR-QC-OM-001	Rev. 00	Email: sales@linesig.com